



Tribhuvan University
Faculty of Humanities and Social Sciences

Project Management System using CPM

A project report

Submitted to
Department of Computer Application
Asian College of Higher Studies

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by
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Supervisor's Recommendation

I hereby recommend that this project prepared under my supervision by Gokarna Chaudhary entitled “**Project Management System using CPM**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

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LETTER OF APPROVAL

This is to certify that this project prepared by Gokarna Chaudhay entitle “**Project Mangement System using CPM**” in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

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ABSTRACT

Efficient project management is critical to ensuring timely task completion and optimal resource utilization. However, many teams continue to rely on manual methods or unstructured tools such as spreadsheets and email, often resulting in missed deadlines, communication breakdowns, and poor task accountability. To overcome these challenges, this project presents the development of "Project Management System using CPM"—a fully implemented web-based project management system integrated with the Critical Path Method (CPM). The system offers features including centralized task assignment, real-time progress tracking, role-based access control, and collaborative task commenting, tailored for small to mid-sized teams. The integration of CPM enables automated identification of critical tasks and dynamic scheduling through visual tools like Gantt charts. Developed using HTML, CSS, JavaScript, PHP, and MySQL, and following the Waterfall model, the system has been rigorously tested to ensure accurate task flow, dependency resolution, and critical path calculations. Results indicate that system determines the project deadline and schedule the tasks.

Keywords: *Project Management System, Web-based Application, Task Scheduling, Critical Path Method (CPM), Task Tracking, Workflow Optimization.*

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Gokarna Chaudhary

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LIST OF ABBREVIATIONS

CPM:	Critical Path Method
CSS:	Cascading Style Sheets
EF:	Early Finish
ER:	Entity-Relationship
ES:	Early Start
FK:	Foreign Key
LF:	Latest Finish
LS:	Latest Start

CHAPTER 1 INTRODUCTION

1.1 Introduction

Project management plays a crucial role in ensuring that tasks and objectives are completed efficiently, on time, and within scope. In the modern digital workspace, managing complex projects, coordinating team members, allocating resources, and tracking progress can become overwhelming without a centralized system [1]. Many teams face challenges such as missed deadlines, poor communication, unclear responsibilities, and inefficient task scheduling, all of which can hinder productivity and project success.

To address these challenges, I have developed Project Mangement System using CPM aimed at improving the way teams plan, execute, and monitor their work. The system provides a structured platform where users can create and manage multiple projects, assign tasks to team members, monitor task progress, and facilitate seamless collaboration in real time.

One of the key features that distinguishes this system from existing tools is its integration of the Critical Path Method (CPM), a project modeling technique used to identify essential tasks that directly impact project completion time. By implementing CPM, the system helps users prioritize critical tasks and allocate resources more effectively, ensuring that delays are minimized and deadlines are met.

Additionally, system supports role-based access control to ensure that users interact with the system based on their responsibilities, thus improving security and task accountability. Its intuitive user interface and real-time updates further contribute to a collaborative and transparent working environment. The primary goal of this system is to simplify project management while enhancing productivity and coordination among team members.

1.2 Problem Statement

Effective project management is essential for ensuring that teams meet deadlines, utilize resources efficiently, and maintain clear communication. However, in many organizations and teams, project management is still handled manually or through unstructured tools such as spreadsheets, email threads, or isolated task lists. This often leads to a variety of issues, including:

- Missed deadlines due to poor task scheduling and oversight
- Lack of accountability and unclear role distribution

- Ineffective communication among team members
- Difficulty in tracking progress and dependencies across tasks

These challenges become even more significant in collaborative environments where multiple team members are working on interdependent tasks. Without a centralized and structured platform, managing projects becomes chaotic, increasing the risk of project failure.

Project Management System using CPM aims to solve these problems by providing a centralized, web-based system that enables streamlined project tracking, task assignment, team collaboration, and deadline monitoring. By integrating advanced scheduling techniques like the Critical Path Method (CPM), the system also helps in identifying and prioritizing key tasks that influence the timely completion of projects.

1.3 Objectives

The main purpose of the project is mentioned below:

- To design and implement task creation, assignment, and status update modules.
- To integrate Critical Path Method (CPM) to calculate project duration and identify critical tasks.
- To develop user authentication with role-based access control (admin and team member).
- To generate Gantt charts and critical path visualization for task scheduling.

1.4 Scope and Limitations

i. The scopes of our project:

- Provides a centralized platform for project management.
- Allows users to assign tasks to specific team members, update task statuses, and monitor progress through clearly defined indicators.
- Implements the Critical Path Method (CPM) for optimized scheduling. Analyzes task dependencies and automatically identifies the critical path to optimize project scheduling and reduce delays.
- Generates graphical representations such as Gantt charts and progress bars to visualize timelines and task progression.
- Implements secure login and access control based on user, ensuring that users only access features relevant to their responsibilities.

ii. The limitations of our project are:

- The system may not support highly complex enterprise-level project management needs initially.
- Resource allocation and budget tracking features are not included in the initial phase.
- Scalability may be limited based on server resources and database performance.
- Learning curve for users unfamiliar with CPM concepts.

CHAPTER 2 BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Project management has been practiced for as long as humanity inhabited earth. There are many examples in history of challenging projects that were successfully completed, despite all the complexities and uncertainties that could've rendered the project a failure. Many of these projects necessitated an enormous workforce, large scope, many years of work, advanced planning and precise execution. It's not clear exactly what the future holds for project management, but with challenges such as globalization, diminishing resources, and increasing population there is no more fitting vehicle for managing such issues than project management. [2]

project management has become an essential discipline across various industries, enabling organizations to handle complex tasks with greater efficiency and control. With the rise of digital transformation, Project Management System using CPM have emerged as powerful tools for planning, scheduling, collaboration, and monitoring. These systems facilitate real-time communication, task delegation, and progress tracking among team members, regardless of location. Furthermore, the integration of methodologies like the Critical Path Method (CPM) allows for data-driven decision-making and optimized resource utilization. As projects grow in scale and complexity, such systems are critical in ensuring timely delivery, risk mitigation, and successful project outcomes.

2.2 Literature Review

It is now ten years since critical path analysis was introduced into British industry. During that time its scope and usefulness have been substantially increased. This article is a critical review of its development. It shows how it has enabled management planning and control systems to be evolved which embrace costs and resources, in addition to time. Factors influencing the degree of success so far are analysed, and potential areas for further development outlined. The range of industrial applications is illustrated. [3]

Hence proper planning and scheduling of project is important to overcome this problem. A project can be defined as a set of a large number of activities or jobs that are performed in a certain sequence determined logically or technologically and it has to be completed within a specified time and cost whilst meeting the performance standards. The project could be

the development of a software program, the building of a house or an office building, development of a new drug, a marketing campaign for a new product, and many others. [4]

In this study, an algorithm was proposed and developed, which determines critical path, critical activities and project completion duration by using GA, instead of CPM and PERT techniques used for network analysis within the scope of project management. The purpose of using GA was that these algorithms are an effective method for solution of complex optimization problems. Therefore, correct decisions can be made for implemented project activities by using obtained results. [5]

The method used in this study is the Critical Path Method (CPM), which is a network method that has a series of activity components with the longest total amount of time and shows the fastest time period of completion. The results of this study indicate that the use of CPMs is able to get the longest total amount of time with the fastest project completion period, project scheduling and project critical paths can be seen more clearly so that the project can be completed on time. [6]

CHAPTER 3 SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

The system analysis of the Project Management System using CPM was carried out using a structured approach aligned with the Waterfall development model. This approach allowed the system to be broken down into well-defined modules such as user management, task handling, and project scheduling. Tools like Data Flow Diagrams (DFDs) and Entity-Relationship (ER) diagrams were used to model the flow of data and relationships between system components. These visual tools helped clearly define the system's processes, inputs, and outputs, enabling organized development and easier implementation. The structured methodology ensured that all functional requirements were thoroughly analyzed, documented, and addressed in a systematic and sequential manner.

3.1.1 Requirement Analysis

i. Functional Requirements:

- **User Roles:**

- Admin: Creates projects, assigns tasks, manages users, re-assign tasks, comment.
- Team Members: Work on assigned tasks, update progress, comment, and communicate.

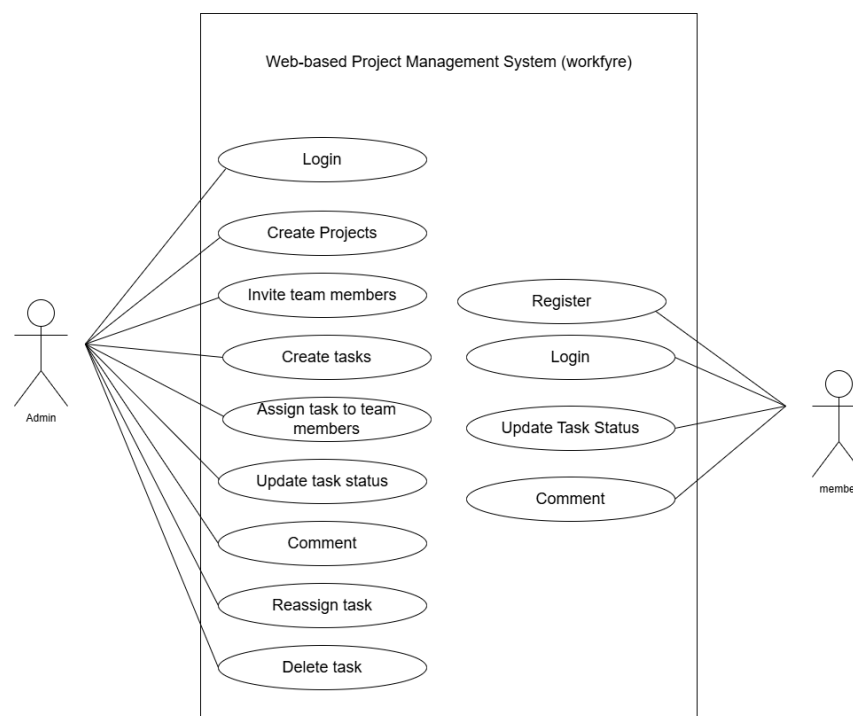


Figure 3.1.1 Usecase Diagram of Project Mangement System using CPM.

- **Core Features:**

- User authentication (login, signup, roles management).
- Project creation and project management.
- Task creation and assignment to team members.
- Task status tracking (Not Started, In Progress, Completed).
- Task dependencies and priority scheduling.
- Comments and notifications for collaboration.
- Gantt charts and CPM-based scheduling.

ii. **Non-Functional Requirements**

- **Usability**

The user interface should be simple and intuitive for both admins and team members.

- **Performance**

The system should respond to user actions (like form submissions or updates) within 2 seconds under normal conditions.

- **Scalability**

The system should support multiple projects and users without significant performance degradation.

- **Security**

Passwords must be securely stored (e.g., using hashing), and unauthorized access to data should be prevented.

- **Reliability**

The system should function consistently without crashing or data loss during typical operations.

- **Maintainability**

The codebase should be modular and documented to allow future updates or feature additions.

- **Compatibility**

The system should work correctly on major modern browsers (e.g., Chrome, Firefox, Edge).

3.1.2 Feasibility Analysis

- **Technical Feasibility:**

System will be developed using a tech stack consisting of:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP
- **Database:** MySQL

- **Economic Feasibility:**

- System will be built using open-source technologies, minimizing development costs.
- Free-tier hosting options make it affordable for small businesses.
- Potential for premium features in future versions for monetization.

- **Operational Feasibility:**

- Designed to be user-friendly with a simple interface.
- Easy onboarding process with minimal learning curve.
- Admins can set up projects and manage users with ease.

3.1.3 Data modelling: ER Diagram

An Entity-Relationship (ER) diagram is a graphical representation used in database design to illustrate the logical structure and relationships among entities within a system. It comprises entities, which represent distinct objects or concepts, and relationships, depicting the associations between these entities. Each entity is defined by attributes, describing its characteristics, while relationships specify how entities are connected. In our case, we have identified User, admin, doctor and schedule register as four entities.

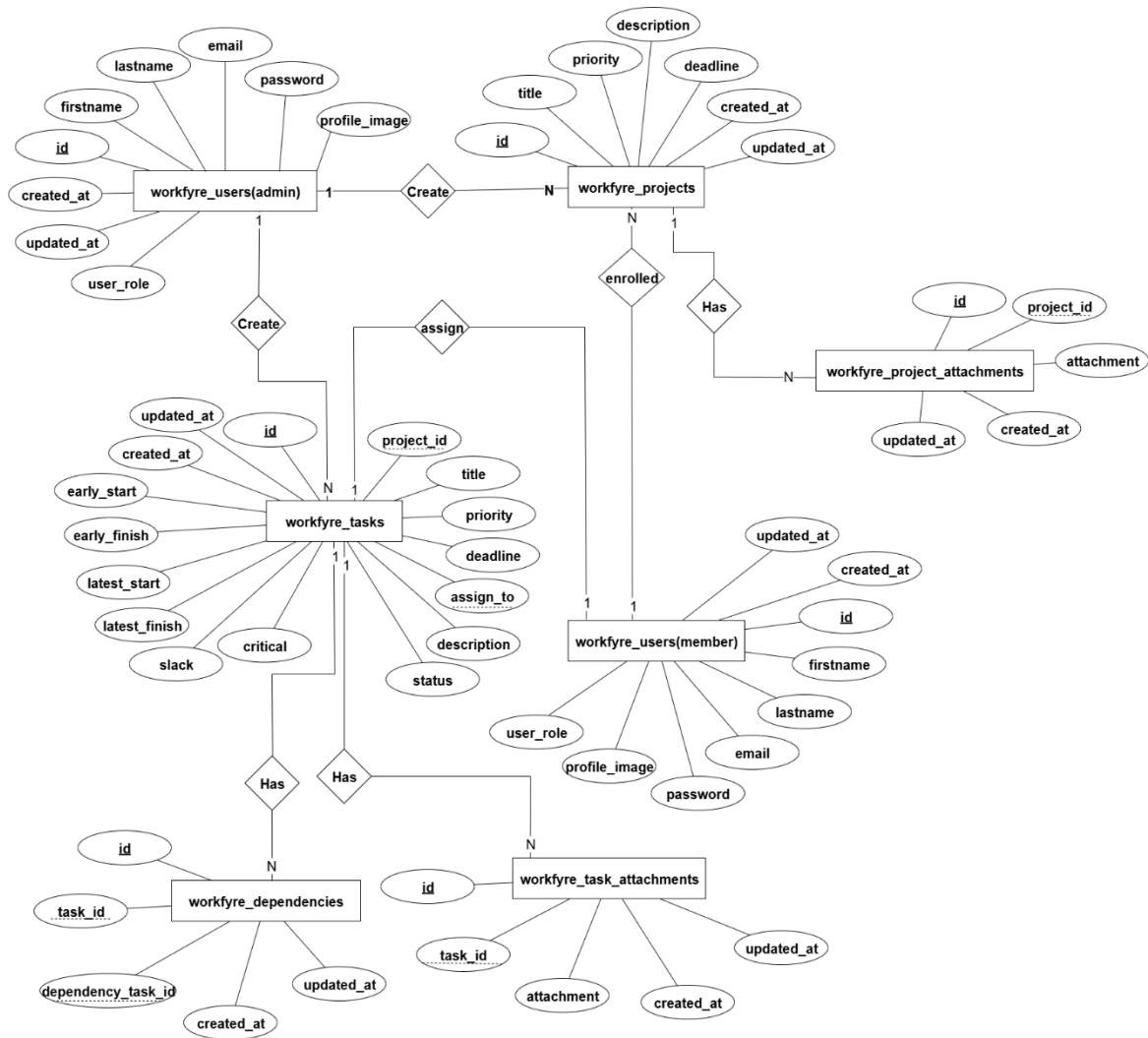


Figure 3.1.3 Er-Diagram of Project Mangement System using CPM

In the above Er-diagram of Project Mangement System using CPM, there are seven entities (workfyre_users, workfyre_projects, workfyre_tasks, workfyre_dependencies, workfyre_task_attachments, workfyre_project_attachments, workfyre_project_meta).

- **Relationships**

- **Users create Projects (Create):** A user can create multiple projects.
- **Projects have Tasks (Has):** A project can have many tasks.
- **Tasks are assigned to Users (Assign):** Each task is assigned to a specific user.
- **Tasks can have Dependencies:** A task can depend on other tasks.
- **Projects have Attachments:** Files or documents related to a project.
- **Tasks have Attachments:** Files or documents related to tasks.
- **Project Meta:** Links users to projects with specific statuses for collaboration or tracking contributions.

3.1.4 Process Modelling: DFD

i. DFD Level 0

The illustration presents the main process in a single node to introduce the project context. This context explains how the project works in just one look. The user feeds the input into the system and then receives the output from it.

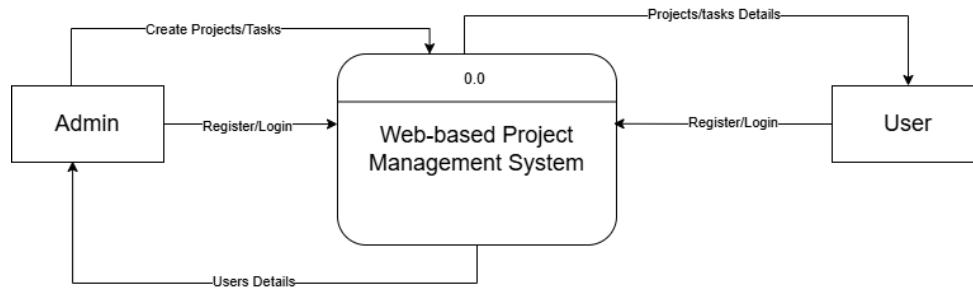


Figure 3.1.4 Level 0 DFD of Project Mangement System using CPM

The Level 0 Data Flow Diagram (DFD) for the Web-based Project Management System provides a high-level overview of how data flows between the main entities and the system. There are two external entities: Admin and User. The admin interacts with the system by logging in and retrieving Users Details, while the User can register or login and submit or retrieve Project Details. The central process, labeled 0.0, represents the core system that manages authentication, user data, and project information. This diagram outlines the basic interactions and data exchange without detailing internal processes or data stores.

ii. Level 1 DFD

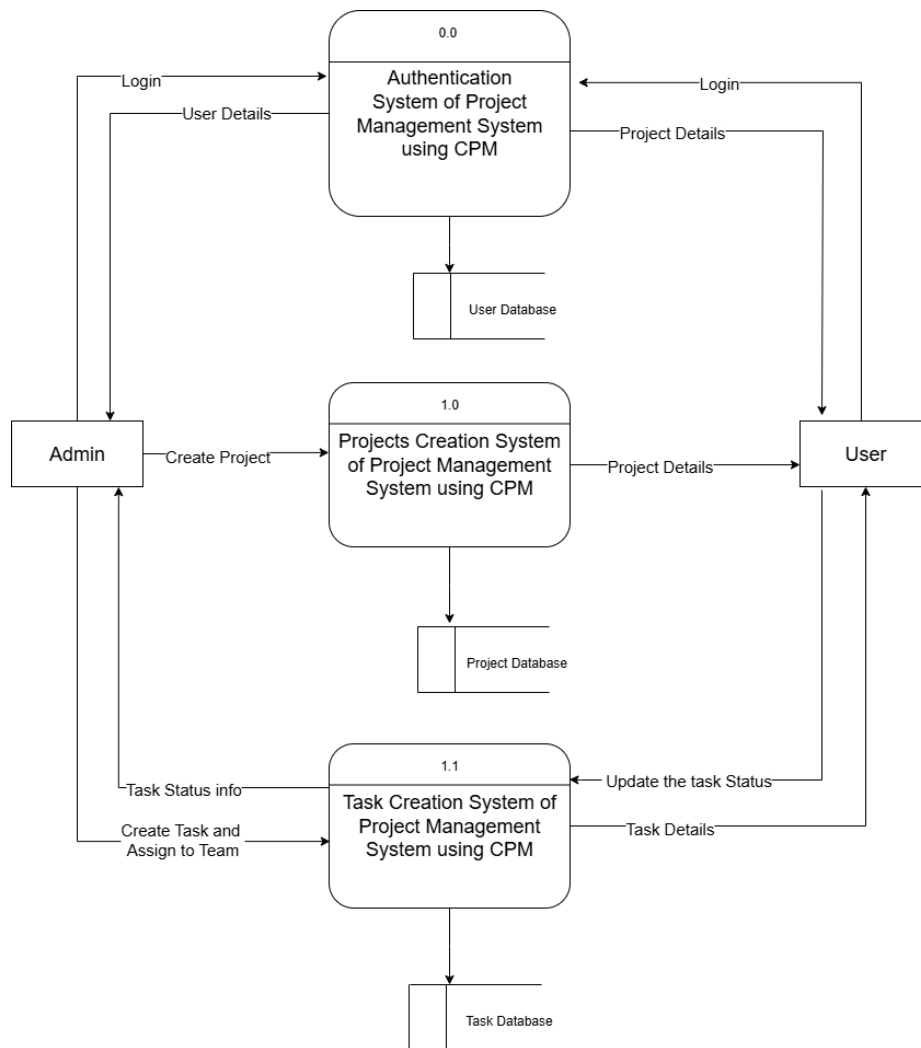


Figure 3.1.5 Level 1 DFD of Project Mangement System using CPM

In the Above level 1 DFD diagram, show the data flow in Web-based Project Mangement System. Admin create projects, invite team to the project. Create task and assign the task to the team. Team members update the task status which they are assigned.

3.2 System Design

The system design of the Project Management System using CPM follows a structured approach using a three-tier architecture comprising the frontend (UI), backend logic, and MySQL database. The architectural design ensures modularity and scalability. The database schema includes normalized tables for users, projects, tasks, and dependencies, supporting accurate CPM calculations. The UI/UX design emphasizes simplicity, with dashboards, task forms, and real-time updates for smooth user interaction. A physical DFD

illustrates actual data flow between users and system processes, ensuring clarity in system operations and integration.

3.2.1 Architecture Design

The architecture of the Project Mangement System using CPM follows a Three-Tier Web Architecture, which ensures scalability, modularity, and separation of concerns. The system is designed to support multiple users (admins and team members) with real-time collaboration and dynamic project scheduling using the Critical Path Method (CPM).

The system design of our Project Mangement System using CPM involves several components and considerations to ensure a seamless user experience and efficient functionality.

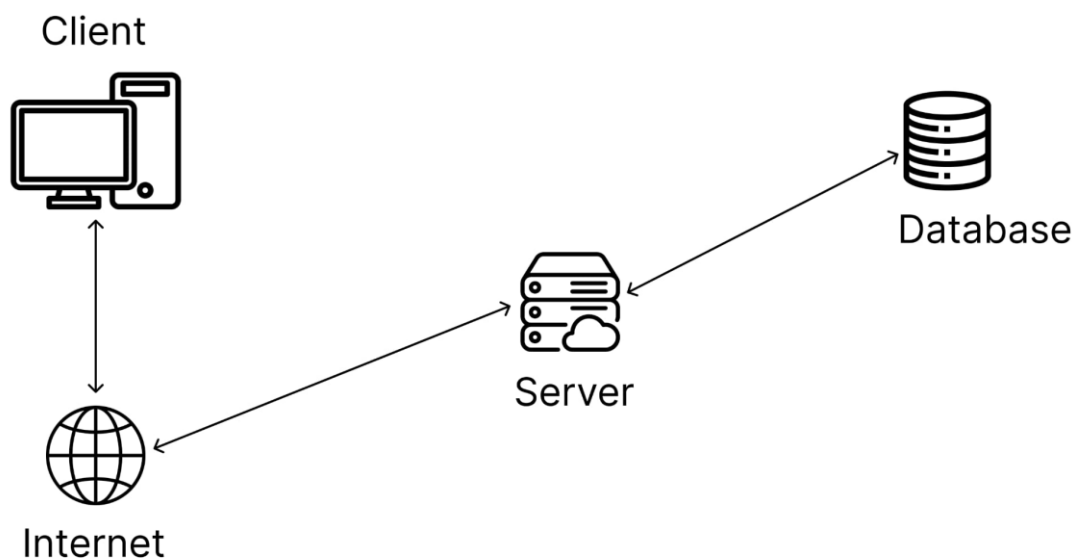


Figure 3.2.1 Three Tier Architecture Daigram of Project Mangement System using CPM

3.2.2 Database Schema Design

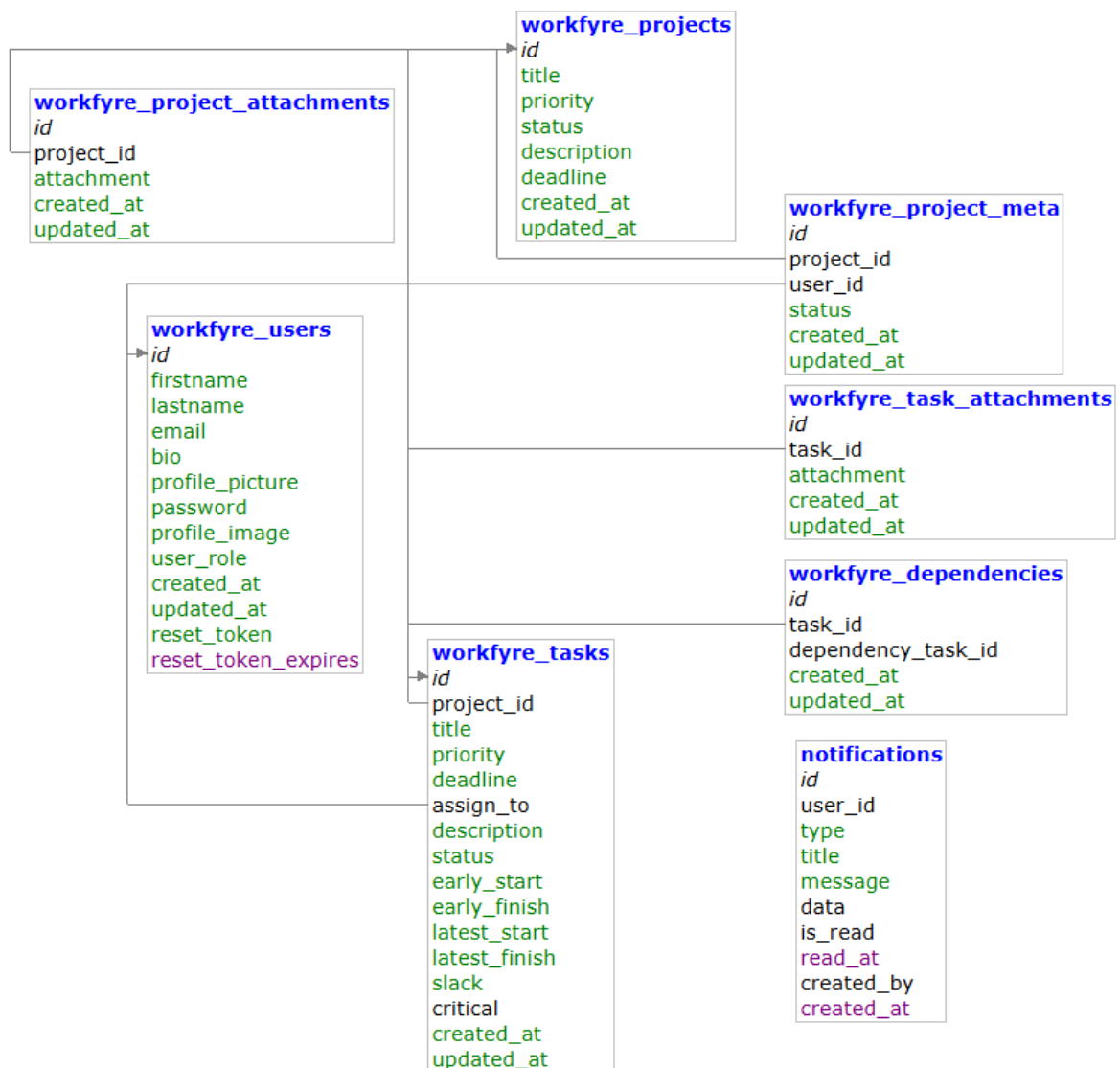


Figure 3.2.2 Database Schema Design of Project Mangement System using CPM

In above database schema design, there are seven tables.

- workfyre_users
 - id (PK): Unique identifier for each user.
- workfyre_projects
 - id (PK): Unique project identifier.
- workfyre_tasks
 - id (PK): Unique task identifier.
 - Foreign Keys:

- `project_id`: References `workfyre_projects(id)` to associate the task with a project.
 - `assign_to`: References `workfyre_users(id)` to assign tasks to users.
- `workfyre_dependencies`
 - Represents task dependencies for CPM analysis.
 - `id` (PK): Unique identifier.
- `workfyre_project_meta`
 - Holds metadata linking users and projects (e.g., assigned members).
 - `id` (PK): Unique identifier.
 - Foreign Keys:
 - `project_id`: References `workfyre_projects(id)`
 - `user_id`: References `workfyre_users(id)`
- `workfyre_project_attachments`
 - Handles file uploads related to projects.
 - `id` (PK): Unique identifier.
 - Foreign Key:
 - `project_id`: References `workfyre_projects(id)`
- `workfyre_task_attachments`
 - Handles file uploads related to tasks.
 - `id` (PK): Unique identifier.
 - Foreign Key:
 - `task_id`: References `workfyre_tasks(id)`

3.2.3 Interface design (UI/UX)

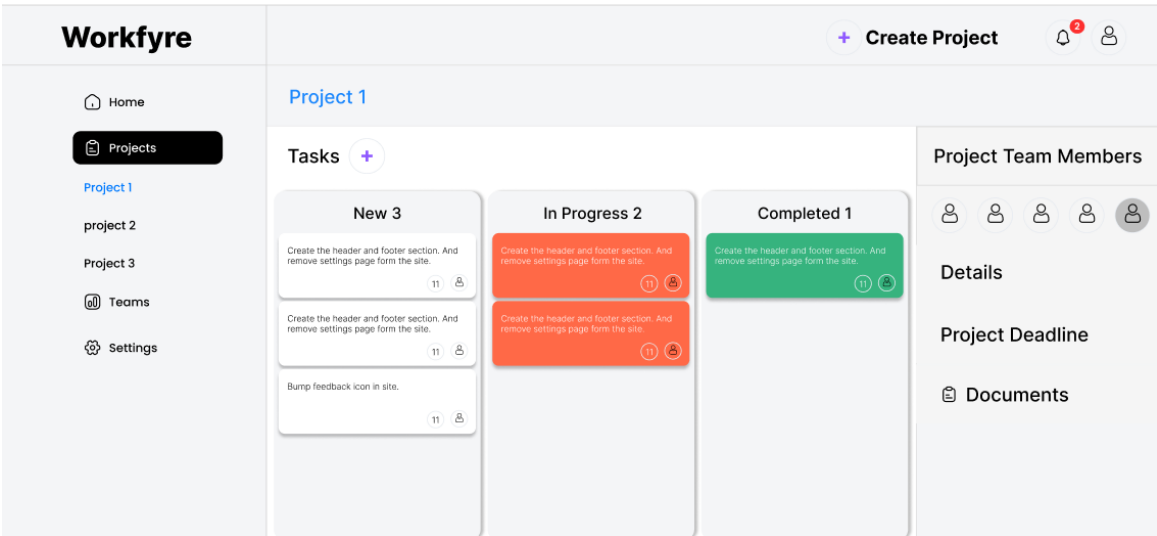


Figure 3.2.3 Interface-Dashboard page

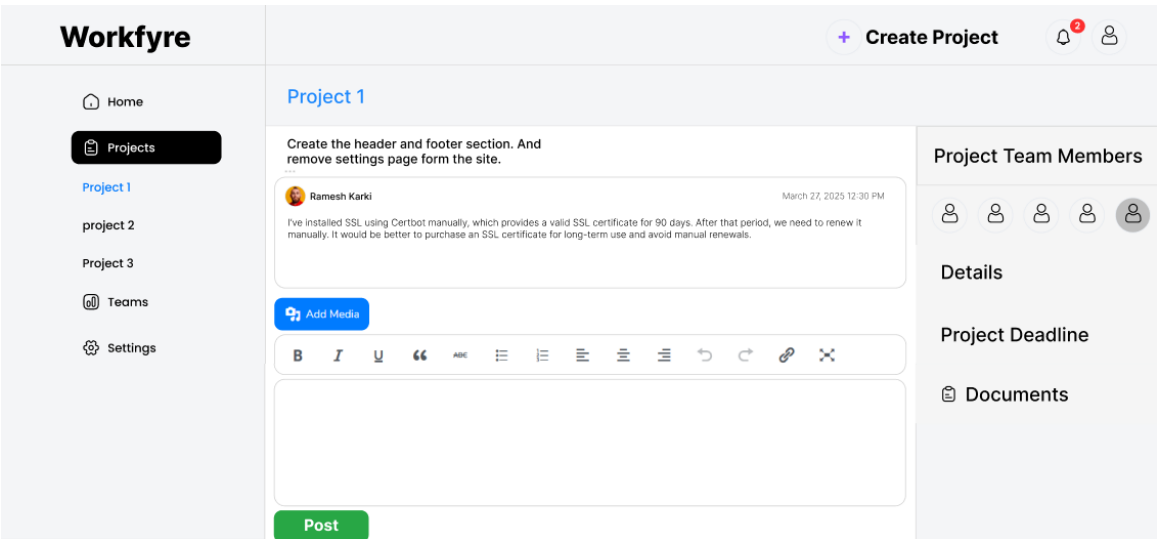


Figure 3.2.4 Interface-comments page

3.2.4 Physical DFD

The Physical Data Flow Diagram (Physical DFD) represents the actual implementation of the Project Management System using CPM. It illustrates how data flows through the system, highlighting specific technologies, user interfaces, and storage components. In this system, users interact with the application through a web browser, submitting data via HTML forms. These inputs are processed by PHP scripts on the server, which perform operations such as login authentication, task assignment, and CPM calculation. Data is

stored and retrieved from a MySQL database, with tables for users, projects, tasks, dependencies, and comments. Notifications and task updates are generated dynamically and displayed through the user interface. The physical DFD helps visualize the real-world execution of system processes, bridging the gap between design and implementation.

3.3 Algorithm Details: Critical Path Method (CPM)

The critical path method (CPM) is a technique that allows you to identify tasks that are necessary for project completion. The critical path in project management is the longest sequence of activities that must be finished on time to complete the entire project. Below, we break down the steps of how you can find the critical path for your next project. It helps you break down complex projects into individual tasks and gain a better understanding of the overall project flexibility.

i. How CPM Will Work in Project Management System using CPM:

- **Task Creation & Dependencies Setup**

- When an admin or project manager creates a project, they define the tasks involved.
- Each task has attributes like:
 - Task Name
 - Duration (Estimated Time to Complete)
 - Dependencies (Tasks that must be completed before this task starts)
 - Assigned Team Members
 - Description
 - Attachments

- **Forward Pass Calculation**

Determines the earliest times at which each task can start and finish.

- If there are no dependencies,
 - $\text{early_start} = 0$
 - $\text{early_finish} = \text{early_start} + \text{duration}$
- If there are dependencies,
 - $\text{early_start} = \text{maximum early_finish of all dependency tasks}$
 - $\text{early_finish} = \text{early_start} + \text{duration}$

This step populates the earliest possible timeline from project start to finish.

- **Backward Pass Calculation**

Determines the latest times at which each task can start and finish without delaying the project.

- If it's the **last task**,
 - $\text{latest_finish} = \text{early_finish}$ (from forward pass)
 - $\text{latest_start} = \text{latest_finish} - \text{duration}$
- For preceding tasks,
 - $\text{latest_finish} = \text{minimum latest_start of all dependent tasks}$
 - $\text{latest_start} = \text{latest_finish} - \text{duration}$

This gives the latest times a task can be done without delaying the project.

- **Slack Calculation**

- $\text{Slack (or Float)} = \text{latest_start} - \text{early_start}$
 - If **Slack = 0**, the task is **critical** (no delay allowed).
 - If **Slack > 0**, the task has **buffer time** and is non-critical.

Slack is stored in the slack column in the workfyre_tasks table.

- **Identifying the Critical Path**

Once early/late times and slack are calculated:

- The system identifies the **Critical Path** as the sequence of tasks where:
 - $\text{Slack} = 0$
 - All tasks are connected via dependencies
- These tasks determine the minimum time required to complete the project.

Stored in the critical field (true/false) in the workfyre_tasks table.

- **Visualization & Monitoring**

A Gantt Chart or Network Diagram is generated to visualize task flow and critical paths.

- As users update task status (status column), the system:
 - Recalculates early/late dates and critical path
 - Updates deadlines dynamically to reflect changes in real-time
- Project managers can see:

- Task progress
- Bottlenecks or delays
- Tasks that need immediate attention (critical)

- **How Project Deadline is Determined**

Once all tasks and dependencies are defined:

- The Critical Path is calculated.
- The sum of durations of tasks on the critical path gives the minimum time required to complete the project.
- The finish time of the last critical task becomes the project's earliest possible deadline.

CHAPTER 4 IMPLEMENTATION AND TESTING

4.1 Implementation

The implementation phase involved the practical realization of the system design into a working web application. System was developed as a full-stack web-based project management tool with integrated Critical Path Method (CPM) calculations for dynamic project scheduling. This chapter describes the development environment, technologies used, module integration, and features implemented in the system.

4.1.1 Tools Used

- **Frontend:** HTML, CSS, JavaScript, Bootstrap
- **Backend:** PHP (Core PHP)
- **Database:** MySQL
- **Tools:** XAMPP (Apache & MySQL), Visual Studio Code
- **Version Control:** Git & GitHub
- **Gantt Chart Library (Optional):** Chart.js / Google Charts / Custom HTML-CSS

4.1.2 Implementation Details of Modules

This section provides detailed implementation insights into the core modules of the Project Management System using CPM, including algorithms, project creation, and task management functionality.

- **Critical Path Algorithm Module**

The critical path algorithm is implemented to automatically identify the longest sequence of dependent tasks that determines the minimum project duration.

Core Function: **calculateCriticalPath(\$taskCriticalDetails)**

```

function calculateCriticalPath($taskCriticalDetails) {
    $criticalPath = [];
    $earliestStart = [];
    $latestFinish = [];

    // Forward pass - calculate earliest start times
    foreach ($taskCriticalDetails as $task) {
        $earliestStart[$task['id']] = 0;
        if (!empty($task['predecessors'])) {
            $maxPredecessorFinish = 0;
            foreach ($task['predecessors'] as $predId) {
                $predFinish = $earliestStart[$predId] + $taskCriticalDetails[$predId]['duration'];
                $maxPredecessorFinish = max($maxPredecessorFinish, $predFinish);
            }
            $earliestStart[$task['id']] = $maxPredecessorFinish;
        }
    }

    // Backward pass - calculate latest finish times
    $projectDuration = max(array_map(function($task) use ($earliestStart) {
        return $earliestStart[$task['id']] + $task['duration'];
    }, $taskCriticalDetails));

    foreach (array_reverse($taskCriticalDetails) as $task) {
        $latestFinish[$task['id']] = $projectDuration;
        if (!empty($task['successors'])) {
            $minSuccessorStart = PHP_INT_MAX;
            foreach ($task['successors'] as $succId) {
                $minSuccessorStart = min($minSuccessorStart, $latestFinish[$succId] - $taskCriticalDetails[$succId]['duration']);
            }
            $latestFinish[$task['id']] = $minSuccessorStart;
        }
    }

    // Identify critical path
    foreach ($taskCriticalDetails as $task) {
        $slack = $latestFinish[$task['id']] - $earliestStart[$task['id']] - $task['duration'];
        if ($slack == 0) {
            $criticalPath[] = $task['id'];
        }
    }
    return $criticalPath;
}

```

- **Forward Pass:** Calculates earliest start times for all tasks
- **Backward Pass:** Calculates latest finish times and slack
- **Critical Path Identification:** Identifies tasks with zero slack
- **Dependency Handling:** Manages complex task dependencies
- **Performance Optimization:** Efficient $O(n^2)$ algorithm for project scheduling

- **Project Creation Module**

The project creation module handles the complete lifecycle of project creation, including validation, database operations, and file management.

Core Function: **createProject(\$params)**

```

function createProject($params) {
    try {
        global $conn;
        $table_name = PREFIX . "projects";

        // Database insertion with prepared statements
        $stmt = $conn->prepare("INSERT INTO $table_name (title, priority, description, created_at, updated_at)
        VALUES (:title, :priority, :description, :created_at, :updated_at)");

        $stmt->bindParam(':title', $params['project_title'], PDO::PARAM_STR);
        $stmt->bindParam(':priority', $params['project_priority'], PDO::PARAM_STR);
        $stmt->bindParam(':description', $params['project_description'], PDO::PARAM_STR);
        $stmt->bindParam(':created_at', $params['created_at'], PDO::PARAM_STR);
        $stmt->bindParam(':updated_at', $params['updated_at'], PDO::PARAM_STR);

        if ($stmt->execute()) {
            $project_id = $conn->lastInsertId();
            return ['status' => 'success', 'project_id' => $project_id];
        }

    } catch (PDOException $e) {
        error_log("Database error: " . $e->getMessage());
        return "Database error: " . $e->getMessage();
    } catch (Exception $e) {
        error_log("An error occurred: " . $e->getMessage());
        return "An error occurred: " . $e->getMessage();
    }
}

// Project creation form handler
$('#createProjectForm').submit(function (e) {
    e.preventDefault();

    var formData = new FormData();
    formData.append('project_title', $('#projectTitle').val());
    formData.append('project_priority', $('#projectPriority').val());
    formData.append('project_description', $('#projectDescription').val());
    formData.append('action', 'create_project');

    // Handle file uploads
    var fileInput = $(this).find('input[name="project_attachments[]"]')[0];
    if (fileInput && fileInput.files.length > 0) {
        for (var i = 0; i < fileInput.files.length; i++) {
            formData.append('project_attachments[]', fileInput.files[i]);
        }
    }

    $.ajax({
        type: 'POST',
        url: 'ajax-project.php',
        data: formData,
        processData: false,
        contentType: false,
        success: function (response) {
            if (response.status == 'success') {
                showSuccessMessage(response.message);
                location.reload();
            } else {
                showErrorMessage(response.message);
            }
        }
    });
});
});

```

- **Input Validation:** Comprehensive form validation
- **File Upload:** Support for multiple file attachments
- **Database Security:** Prepared statements prevent SQL injection
- **Error Handling:** Comprehensive error logging and user feedback
- **AJAX Integration:** Seamless user experience with asynchronous requests

- **Task Creation Module**

The task creation module manages task lifecycle, including creation, assignment, dependency management.

Core Function: **createTask(\$params)**

```
function createTask($params) {
    try {
        global $conn;
        $table_name = PREFIX . "tasks";
        $created_by = $_SESSION['user_id'] ?? null;

        // Set default priority if not provided
        $priority = isset($params['task_priority']) ? $params['task_priority'] : 'low';

        $stmt = $conn->prepare("INSERT INTO $table_name (project_id, title, priority, deadline, assign_to, description,
        | created_at, updated_at)
        VALUES (:project_id, :title, :priority, :deadline, :assign_to, :description, :created_at, :updated_at)");

        $stmt->bindParam(':project_id', $params['project_id'], PDO::PARAM_INT);
        $stmt->bindParam(':title', $params['task_title'], PDO::PARAM_STR);
        $stmt->bindParam(':priority', $priority, PDO::PARAM_STR);
        $stmt->bindParam(':deadline', $params['task_deadline'], PDO::PARAM_STR);
        $stmt->bindParam(':assign_to', $params['task_assignto'], PDO::PARAM_INT);
        $stmt->bindParam(':description', $params['task_description'], PDO::PARAM_STR);
        $stmt->bindParam(':created_at', $params['created_at'], PDO::PARAM_STR);
        $stmt->bindParam(':updated_at', $params['updated_at'], PDO::PARAM_STR);

        if ($stmt->execute()) {
            $task_id = $conn->lastInsertId();

            // Create comprehensive notifications
            if ($task_id && $created_by) {
                $notification_result = createTaskCreationNotificationComprehensive(
                    $task_id,
                    $created_by,
                    $params['task_assignto'] ?? null
                );
            }
            return ['status' => 'success', 'task_id' => $task_id];
        }
    } catch (PDOException $e) {
        error_log("Database error in createTask: " . $e->getMessage());
        return "Database error: " . $e->getMessage();
    } catch (Exception $e) {
        error_log("An error occurred in createTask: " . $e->getMessage());
        return "An error occurred: " . $e->getMessage();
    }
}
```



```
// Automatic critical path analysis after task creation
function updateTaskCriticalPathParams($params) {
    try {
        global $conn;
        $table_name = PREFIX . "tasks";

        $stmt = $conn->prepare("UPDATE $table_name SET
            early_start = :early_start,
            early_finish = :early_finish,
            latest_start = :latest_start,
            latest_finish = :latest_finish,
            slack = :slack,
            critical = :critical
            WHERE id = :task_id");

        $stmt->bindParam(':early_start', $params['early_start'], PDO::PARAM_STR);
        $stmt->bindParam(':early_finish', $params['early_finish'], PDO::PARAM_STR);
        $stmt->bindParam(':latest_start', $params['latest_start'], PDO::PARAM_STR);
        $stmt->bindParam(':latest_finish', $params['latest_finish'], PDO::PARAM_STR);
        $stmt->bindParam(':slack', $params['slack'], PDO::PARAM_STR);
        $stmt->bindParam(':critical', $params['critical'], PDO::PARAM_INT);
        $stmt->bindParam(':task_id', $params['task_id'], PDO::PARAM_INT);

        return $stmt->execute();
    } catch (Exception $e) {
        error_log("Error updating critical path params: " . $e->getMessage());
        return false;
    }
}
```

- **Task Assignment:** Automatic user assignment with validation
- **Dependency Management:** Support for task dependencies
- **Critical Path Analysis:** Automatic critical path calculation
- **Notification System:** Comprehensive notification delivery
- **File Attachments:** Support for task-related files
- **Priority Management:** Automatic priority assignment based on critical path

4.2 Testing

This testing report provides a comprehensive analysis of the project management system using CPM, covering both unit testing and system testing phases. The system is a PHP-based web application that handles user authentication, project management, task creation, and team collaboration features.

4.2.1 Test cases for Unit Testing

- Authentication Functions Testing

Module	Function	Test Case	Expected Results
Authentication Functions	registerUser	- Valid Registration - Email Duplicate Check - Password Hashing - Session Creation - Database Error Handling - Input Validation	- Success: returns true and creates user session - Failure: returns error message string
	loginUser	- Valid Login - Invalid Password - Non-existent User - Password Verification - Session Management - Database Error Handling	- Success: returns true and creates user session - Failure: returns error message ("Invalid password" or "User not found")

Table 4.2.1 Authentication unit test case of project management using CPM

- Project Management Functions Testing

Module	Function	Test Case	Expected Results
Project Management	createProject	- Valid Project Creation - Database Insertion - Project ID Return - Required Fields - Database Error Handling - Data Validation	- Success: Returns ['status' => 'success', 'project_id' => \$id] - Failure: Returns error mess
	Project Retrieval Functions (getProjectDetails, getProjectById, getProjectCount)	- Retrieve project list with pagination - Retrieve specific project by ID - Get total project count - Handle empty results - Invalid parameters (limit/offset) handling	- Success: Returns correct project data - Failure: Handles empty or invalid inputs gracefully
	createTask	- Valid Task Creation - Default Priority assignment - Task assignment to users - Notification creation - Critical path detection - Database error handling - Required fields	- Success: Returns ['status' => 'success', 'task_id' => \$id] - Failure: Returns error message string

Table 1 Project Management unit test case of project management using CPM

4.2.2 Test cases for System Testing

- Authentication System Testing

Module	Test Scenario	Test Case	Expected Results
Authentication System	User Registration Flow	- Valid Registration - Invalid Data (e.g. email format, names with numbers) - Duplicate Email - Password Mismatch - Network Issues	- Form validates inputs - AJAX request succeeds - User created in DB - Session created - Redirect to dashboard
	User Login Flow	- Valid Login - Invalid Password - Non-existent User - Invalid Email Format - Session Persistence	- Email validated - Password verified - Session created - User redirected to dashboard

- Project Management System Testing

Module	Test Scenario	Test Case	Expected Results
Project Management	Project Creation	- Valid Project Creation - File Upload (types/sizes) - Missing Required Fields - Large File Upload - Invalid File Types	- Valid data accepted - Project saved - Files uploaded - New project shown in list
	Task Creation	- Valid Task Creation - Task Assignment	- Task saved - Assigned correctly

		- Task Dependencies - Critical Path Detection - File Attachments - Notification System	- Files uploaded - Notifications sent - Path calculated
	Task Operations	- Status Updates (pending → in-progress → completed) - Priority Changes - Reassignment - Manage Dependencies - Task Comments - Task History	- Updates reflected in UI - History and dependencies logged - Comments stored and viewable

- Team Collaboration System Testing

Module	Test Scenario	Test Case	Expected Results
Project Management	Team Invitation	- Invitations Sent Successfully - Valid Invitation Link in Email - Accept Invitation Functionality	- Emails sent with correct links - Users can join project - Members added with appropriate roles
	Team Management	- Add New Team Member - Assign Roles (admin/member) - Permission Level Testing	- Team members managed correctly - Role and permission logic enforced

- CPM Algorithm Testing

ID	Test Scenario	Input Data	Expected Results	Status
1	Create all tasks with duration and dependencies	Tasks: A (2), B (4) → A, C (5) → B, D (4) → B, E (2) → C, D, F (3) → E	Tasks saved with correct dependencies and durations	Pass
2	Calculate earliest start and finish times	and finish times Based on input task graph	Correct early start and early finish: A: 0–2, B: 2–6, C: 6–11, D: 6–10, E: 11–13, F: 13–16	Pass
3	Calculate latest start and finish times	Use total project duration from forward pass	Correct latest start and finish: A: 0–2, B: 2–6, C: 6–11, D: 7–11, E: 11–13, F: 13–16	Pass
4	Calculate slack (float) for all tasks	Use early and latest start/finish times	Tasks A, B, C, E, F have 0 slack; D has 1 day slack	pass
5	Determine critical path from tasks	Input task graph with calculated early/late times	Critical path identified as: A → B → C → E → F	pass
6	Calculate total project time	Based on critical path	Total project duration = 16 days	pass
7	Delay a non-critical task (D) by 1 day	Delay D's duration or start by 1 day	Project duration remains 16 days, slack consumed	pass
8	Delay a critical task (e.g., C)	Delay C start by 1 day	Total project duration increases to 17 days; critical path impacted	pass

Table 4.2.9 Test cases of critical path method algorithm in workfyre

CHAPTER 5 CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 conclusion

The development of the Project Management System using CPM has successfully addressed key challenges in project planning and team collaboration by implementing features such as task assignment, role-based access, and the Critical Path Method (CPM) for effective scheduling. All core functionalities—including project creation, task management, task commenting, notifications, responsive design, and CPM integration—have been fully implemented. The system is now feature-complete and ready for deployment, offering a streamlined, web-based solution tailored for small to mid-sized teams. By combining practical project management tools with dynamic scheduling, the system enhances overall efficiency, accountability, and user experience.

5.2 Lesson Learnt

Developing the Project Management System using CPM offered valuable insights into both technical and non-technical dimensions of software development. Throughout the phases of planning, analysis, design, and implementation, several important lessons emerged:

- Clear requirement gathering is essential to avoid delays and costly rework.
- Implementing CPM deepened understanding of task dependencies, durations, and project timelines.
- CPM integration highlighted the importance of accurate task data and validation logic.
- Applying project management principles improved scheduling and task sequencing awareness.
- Iterative development helped refine the system, incorporate feedback, and meet deadlines.
- Rigorous testing, including unit and integration tests, is crucial for system stability and reliability.

5.3 Future Recommendations

While the current version of the Project Management System using CPM successfully delivers its core functionalities, there are several areas that can be further enhanced in future iterations to increase usability, scalability, and overall impact:

- **Integration with Third-Party Tools**

Incorporating integration with tools like Google Calendar, Slack, or Microsoft Teams can improve team collaboration and real-time updates. This would allow users to sync tasks, deadlines, and receive notifications on their preferred platforms.

- **Advanced Analytics and Reporting**

Adding dashboards and data visualizations (e.g., Gantt charts, burn-down charts, performance metrics) can help users gain deeper insights into project progress, resource allocation, and potential bottlenecks.

- **Mobile Application Development**

Developing a mobile version of the system (Android and iOS) would enhance accessibility and allow team members to manage tasks and receive notifications on the go.

- **Role Customization and Permission Granularity**

Introducing customizable roles and more granular permission controls can help better accommodate different organizational structures and security requirements.

- **AI-Based Task Suggestions**

Future versions could include AI-driven features such as automatic task prioritization, estimated completion times based on historical data, or risk alerts based on project trends.

- **Offline Mode and Data Synchronization**

Adding offline functionality would allow users to view and modify projects without an internet connection, with changes syncing automatically once the connection is restored.

- **Scalability and Cloud Deployment**

To support larger teams or organizations, deploying the system on scalable cloud infrastructure (e.g., AWS, Azure) with multi-tenant capabilities would improve performance and reliability.

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APPENDIX

Workfyre

Home

Projects

Teams

Accounts

Notifications

Settings

Logout

Gokarna

gokarnachy

Search here

159

+ New Project

Good Morning, Gokarna!

Welcome back! You have 6 team members and 3 total projects to manage.

Sunday, July 27, 2025

7:26 AM

Total Projects

3

Across all teams

Active Projects

0

Currently in progress

Completed

0

Successfully finished

Completion Rate

0%

Project success rate

Recent Projects

View All

I

Implementation of a New CRM System

Contrary to popular belief, Lorem Ipsum is not sim...

Not started

Jul 28

D

Developing a Sustainable Urban Farming Initiative

Contrary to popular belief, Lorem Ipsum is not sim...

Not started

Jul 28

T

The Impact of Technology on Modern Education

Contrary to popular belief, Lorem Ipsum is not sim...

Not started

Jul 28

Your Tasks

View All

No tasks assigned

Tasks will appear here when assigned to you

Dashboard

Workfyre

Home

Projects

Teams

Accounts

Notifications

Settings

Logout

Gokarna

gokarnachy

Search here

159

+ New Project

Project Dashboard

All Projects

Filter Projects

Clear All Filters

Search Projects

Project Status

Priority

Sort By

Search by title...

All Status

All Priorities

Latest First

High

Implementation of a New CRM System

Progress

40%

10 4

High

Developing a Sustainable Urban Farming Initiative

Progress

0%

0 0

Quick Stats

Total Projects

Active Projects

In Progress

Completed

3

0

0

0

projects page

Workfyre

Home

Projects

Teams

Accounts

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Logout

Gokarna

gokamachy

Search here

199

New Project

Implementation of a New CRM System

Active

Invite Team

Contrary to popular belief, Lorem Ipsum is not simply random text. It has roots in a piece of classical Latin literature...

Progress 40%

10 Total Tasks

4 Completed

24 Days Left

Team Members:

Export

Share

Project Timeline - Gantt Chart

137 figma Desing (2 days) (Non-critical)

138 Frontend Development (4 days) (Non-critical)

139 Responsiveness (7 days) (Non-critical)

140 Backend (8 days) (Non-critical)

444 Backend Development (10 days) (Non-critical)

Single project page

Workfyre

Home

Projects

Teams

Settings

Logout

Search here

Notifications

Settings

Profile

← Logo Design for saino cafe

this is he logo design project...

Responsiveness

Comments

Files

B

/

List

Link

Media

Comment

Single task page

Workfyre

Home

Projects

Teams

Accounts

Notifications

Settings

Logout

Gokarna
gokarnachy

Search here

+39 + New Project

Team Management

Manage your team members and their project assignments

+ Add Team Member

Total Team Members
0



Active Members
0



Project Assignments
0



Members with Projects
0



Filter Team Members

Clear All

Search Members

Status

Sort By

Search by name or email...

All

Latest First

Team Members (6)

Select All

☐ **andy ruth**
heyafax138@7tul.com
1 projects Joined Jul 2025
ENROLLED PROJECTS
Implementation of a New CRM System

Enrolled

Select a team member
Click on a team member to view their details

Teams page

Workfyre

Home

Projects

Teams

Accounts

Notifications

Settings

Logout

Gokarna
gokarnachy

Search here

+39 + New Project

Notification Dashboard

Notification

Stay Updated With Your Latest Notifications

Today

Gokarna Chaudhary
Create Rest api for data access. Now

Gokarna Chaudhary
Create Rest api for data access. 3 min ago

Yesterday

Gokarna Chaudhary
Create Rest api for data access. Feb 12, 2025 2:00 PM.

Gokarna Chaudhary
Create Rest api for data access. Feb 13, 2025 2:30 PM.

Filter By

Mark all as read

☒ All

☐ Seen

☐ Unseen

Notifications page



Profile

Update your photo and personal details.

Profile

Security

First Name

Gokarna

Last Name

Chaudhary

Email

gokarnachy28@gmail.com

Email cannot be changed

Role

Admin

Settings page